EQUIVALENT CHARACTERIZATIONS OF AN ELEMENT OF THE MAXIMAL IDEAL THAT CUTS OUT THE MAXIMAL IDEAL UP TO RADICAL

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Zariski topology criterion for the radical of a principal ideal; spectrum of a localization. Stacks Project, Algebra, Lemma algebra-lemma-Zariski-topology (tag 00IY) and Lemma algebra-lemma-spec-localization (tag 00E4). Give the equivalence of (1) and (2) and the description of Spec(A_f).
- Zero-dimensional rings as products of localizations; associated primes; total ring of fractions. Stacks Project, Algebra, Proposition algebra-proposition-dimension-zero-ring (tag 00JA); Proposition algebra-proposition-minimal-primes-associated-primes (tag 00LD); Lemma algebra-lemma-total-ring-fractions-no-embedded-points (tag 02LX). Supply the identification of A_f with a product of the localizations and the depth-one and total-ring-of-fractions equivalences.

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Lemma 0.1. *Let $(A, \mathfrak{m})$ be a Noetherian 1-dimensional local ring. Let $f \in \mathfrak{m}$. The following are equivalent*

(1) $\mathfrak{m} = \sqrt{(f)}$,
(2) *f is not contained in any minimal prime of A, and*
(3) *$A_f = \prod_{\mathfrak{p}\ minimal} A_{\mathfrak{p}}$ as A-algebras.*

Such an $f \in \mathfrak{m}$ exists. If $\text{depth}(A) = 1$ (for example A is reduced), then (1) – (3) are also equivalent to

(4) *f is a nonzerodivisor,*
(5) *A_f is the total ring of fractions of A.*

If A is reduced, then (1) – (5) are also equivalent to

(6) *A_f is the product of the residue fields at the minimal primes of A.*

Proof. The spectrum of A has finitely many primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_n$ besides $\mathfrak{m}$ and these are all minimal, see Algebra, Lemma 00FR. Then the equivalence of (1) and (2) follows from Algebra, Lemma 00E0. Clearly, (3) implies (2). Conversely, if (2) is true, then the spectrum of A_f is the subset $\{\mathfrak{p}_1, \ldots, \mathfrak{p}_n\}$ of $\text{Spec}(A)$ with induced topology, see Algebra, Lemma 00E3. This is a finite discrete topological space. Hence $A_f = \prod_{\mathfrak{p}\ \text{minimal}} A_{\mathfrak{p}}$ by Algebra, Proposition 00KJ. The existence of an f is asserted in Algebra, Lemma 00KK.

Assume A has depth 1. (This is the maximum by Algebra, Lemma 00LK and holds if A is reduced by Algebra, Lemma 031R.) Then $\mathfrak{m}$ is not an associated prime of A. Every minimal prime of A is an associated prime (Algebra, Proposition 02CE). Hence the set of nonzerodivisors of A is exactly the set of elements not contained in any of the minimal primes by Algebra, Lemma 00LD. Thus (4) is equivalent to (2). Part (5) is equivalent to (3) by Algebra, Lemma 02LX.

Then $A_{\mathfrak{p}}$ is a field for $\mathfrak{p} \subset A$ minimal, see Algebra, Lemma 00EU. Hence (3) is equivalent to (6). □